LECTURE L29-30 · MODULE 4 · CO4

Closing the loop

Week 15 | Slides: `DATA209_Module_4_Feature_Engineering_and_Structure.pptx`, pages 29–37 | Laboratory: **P29-30**

TOPICS COVERED

- Hypothesis refinement; train/test splitting
- Cross-validation overview
- Data leakage: concept and prevention
- Translating EDA into insight

Splitting

The test set exists to estimate performance on data the process has never seen. **Touch it once.**

- Split **before** imputing, scaling, encoding, selecting or transforming — before everything.
- **Stratify** on the target when classes are imbalanced.
- **Time-ordered data** requires a time-based split: train on the past, test on the future, never shuffle.
- **Grouped data** must be split by group, so all records for one entity sit on one side.
- Set and record a random seed so the marker reproduces exactly what you saw.

Cross-validation

One split gives one noisy estimate. k-fold splits into k parts, training on k−1 and testing on the remainder, k times; k = 5 or 10 is standard. **Stratified k-fold** preserves class proportions and is the default for imbalanced classification. **Time-series split** uses expanding or rolling windows. **Group k-fold** keeps entities together.

Report the mean **and the spread** across folds. A high mean with a wide spread is an unstable result, not a good one.

Data leakage

> Leakage is any path by which information from the test set — or from the future — reaches the training process.

It is the most expensive error in applied data science because it announces itself as *excellent* validation performance and only fails in production.

PATH	RISK	FIX
Preprocessing on all data	scaler or imputer sees test statistics	fit inside a `Pipeline`, on train only
Target leakage	a feature contains the answer	audit each feature's availability at prediction time
Temporal leakage	training on rows recorded after test rows	split by time; check both date ranges
Duplicate leakage	the same entity in train and test	deduplicate first, then split by group

The diagnostic question for every feature: *would this value have been available, with this content, at the moment of prediction?* A useful blunt check is single-feature AUC — anything above about 0.95 suggests the feature is standing in for the target.

Sampling skewed data — imbalanced targets

Identify the imbalance during EDA and state it. Accuracy is not honest at 85/15. Report precision, recall, F1 and PR-AUC, and choose by which error costs more. Stratify every split and every fold. Resampling belongs inside the training fold only; class weights are simpler and leak nothing.

Turning exploration into insight

The structure of a credible EDA report, and of your final project:

1. **What the data is** — source, sampling frame, rows, columns, period; what it cannot speak to.
2. **What was wrong with it** — the cleaning log, and every decision you made.
3. **What it shows** — findings, each with the figure that supports it and a sentence of meaning.
4. **What comes next** — hypotheses with falsification conditions, the modelling-ready dataset,

and the caveats.

A negative result, well established, is a contribution. "K-means finds no strong segmentation (silhouette 0.20)" is a finding worth reporting.

Review — the semester in one page

MODULE	THE QUESTION IT ANSWERS	THE ARTEFACT IT PRODUCES
1 · Foundations	what is this data, and what may I claim from it?	a described dataset with its sampling frame stated
2 · Core EDA	what patterns and structure does it contain?	findings and a hypothesis shortlist
3 · Data quality	can I trust it?	a cleaning log and a reliable analytical dataset
4 · Structure	what shape should it take for modelling?	a leak-free, modelling-ready feature matrix

Five ideas that recur in every module, and that the final project is really marking:

1. **Look before you test.** Compute the summary, then plot it, then reconcile the two.
2. **Every choice is an assumption.** Imputing, deleting, transforming and encoding all assert

something. Write the assertion down.

1. **Match the statistic to the variable.** Scale decides the statistic; shape decides the summary.

2. **Measure the effect of your own decisions.** Before-and-after, every time.
3. **Say what the data cannot support.** The limitations section is part of the finding, not an

apology for it.

KEY TERMS

`Train/test split` `Stratification` `k-fold cross-validation` `Group k-fold` `Time-series split` `Data leakage` `Target leakage` `Temporal leakage` `PR-AUC` `Pipeline`

COMMON ERRORS TO AVOID

- Fitting any transformer before the split
- Reporting accuracy on an imbalanced target
- Evaluating on the test set more than once
- Resampling or selecting features outside the cross-validation loop

READINGS — ALL FREE TO ACCESS

1. scikit-learn — common pitfalls and data leakage
   `https://scikit-learn.org/stable/common_pitfalls.html`
2. scikit-learn — cross-validation
   `https://scikit-learn.org/stable/modules/cross_validation.html`
3. Berkeley Data 100 — course notes
   `https://ds100.org/course-notes/`

TEST YOURSELF

1. Name the four leakage paths and one fix for each.
2. Why is a random split wrong for a dataset with timestamps?
3. A feature alone gives AUC 0.98. What is your first suspicion?
4. Give the four-part structure of an EDA report.